

FRANK PORTER
2016-2017

M1 MACRO

Group B

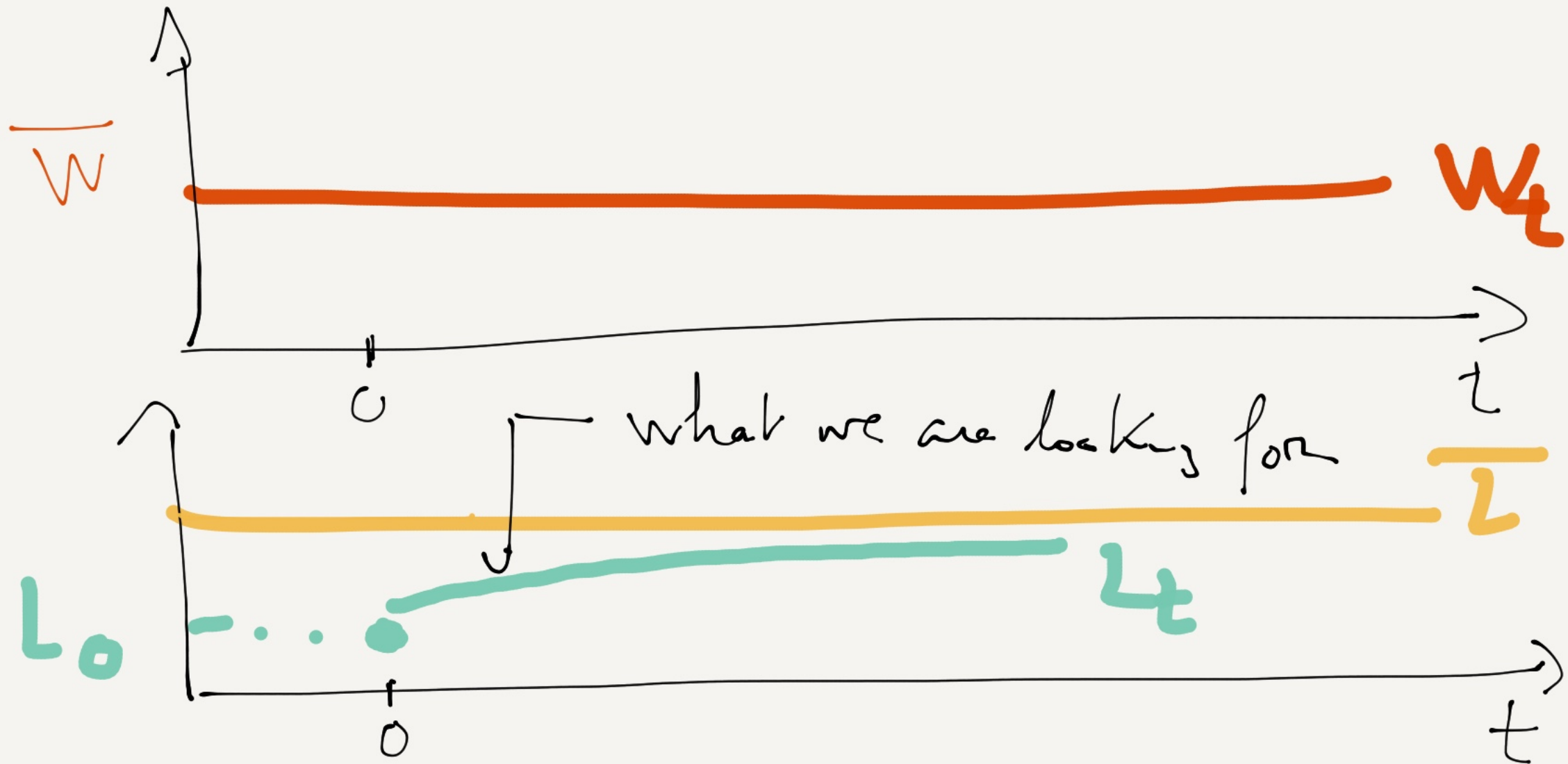
Session 14

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

$$\beta b \bar{L}_{t+1} - (d_1 + b + \beta b) \bar{L}_t + b \bar{L}_{t-1} = 0$$

[Assume $\bar{W}_t = 0$]

Note $\bar{W}_t = 0 \quad \forall t \Rightarrow W_t = \bar{W}$



How to find c_1 and c_2 ?

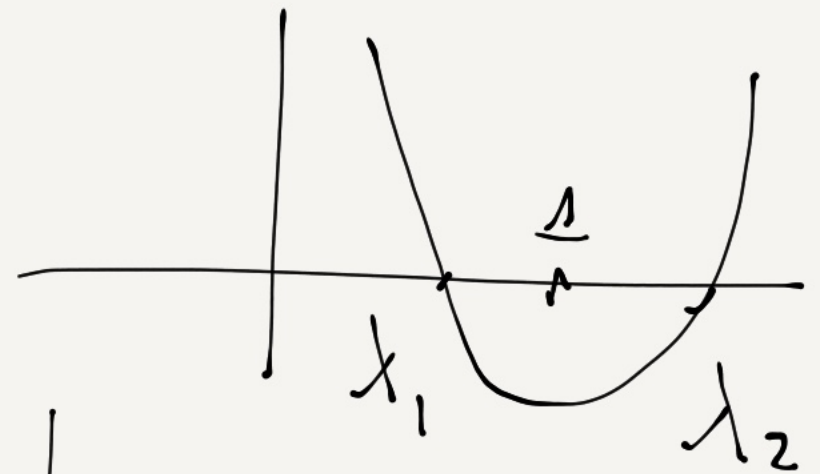
• $\underline{L_0} = c_1 \lambda_1^0 + c_2 \lambda_2^0 = c_1 + c_2 \quad \leftarrow \text{1 eq in } (c_1, c_2)$
Known

• Transversality condition : $\begin{cases} \lim_{t \rightarrow \infty} \bar{L}_t = 0 \\ \Leftrightarrow \lim_{t \rightarrow \infty} L_t = \bar{L} \end{cases}$

$\hookrightarrow \lim_{t \rightarrow \infty} c_1 \lambda_1^t + c_2 \lambda_2^t = c_1 \underbrace{\lim_{t \rightarrow \infty} \lambda_1^t}_0 + \lim_{t \rightarrow \infty} c_2 \lambda_2^t = 0$
if only if $c_2 = 0$

I know that the solution is

$$\hat{L}_t = c_1 \lambda_1^t + c_2 \lambda_2^t$$



where c_1, c_2 have to be found

• λ_1, λ_2 are the roots of $\beta b \lambda^2 - (\alpha_1 + b + \beta b) \lambda + b = 0$

• We have shown that $0 < \lambda_1 < 1$
 $\lambda_2 > 1$

• L_0 is given

$$\Rightarrow \lambda_{1,2} = \frac{(\alpha_1 + b + \beta b) \pm \sqrt{(\alpha_1 + b + \beta b)^2 - 4\beta b^2}}{2\beta b}$$

$$\Rightarrow c_2 = 0, \quad c_1 = \hat{L}_0 = L_0 - \bar{L}$$

Therefore, the optimal path is $\hat{L}_t = \hat{L}_0 \lambda_1^t$

$$= \lambda_1 \times \underbrace{\hat{L}_0 \lambda_1^{t-1}}_{\hat{L}_{t-1}}$$

$$\Rightarrow \boxed{\hat{L}_t = \lambda_1 \hat{L}_{t-1}}$$

$$L_t - \bar{L} = \lambda_1 (L_{t-1} - \bar{L}) \quad \bar{L} = \frac{d_0 - \bar{w}}{\alpha_1}$$

$(\Leftrightarrow)$

$$\boxed{L_t = (1 - \lambda_1) \bar{L} + \lambda_1 L_{t-1}}; L_0 \text{ given}$$

Description of the optimal path

• $w_t = \bar{w} \quad \forall t \text{ from } -\infty \text{ to } +\infty$

• Assume L_0 is given

$$L_1 = (1 - \lambda_1) \bar{L} + \lambda_1 L_0 = \bar{L} - \lambda_1 (\bar{L} - L_0)$$

→ λ_1 is here a decreasing function of b

high $b \Rightarrow$ high adj. costs $\Rightarrow$ slow adjustment

$\Rightarrow \lambda_1$ is close to 1 $\Rightarrow L_1$ close to L_0

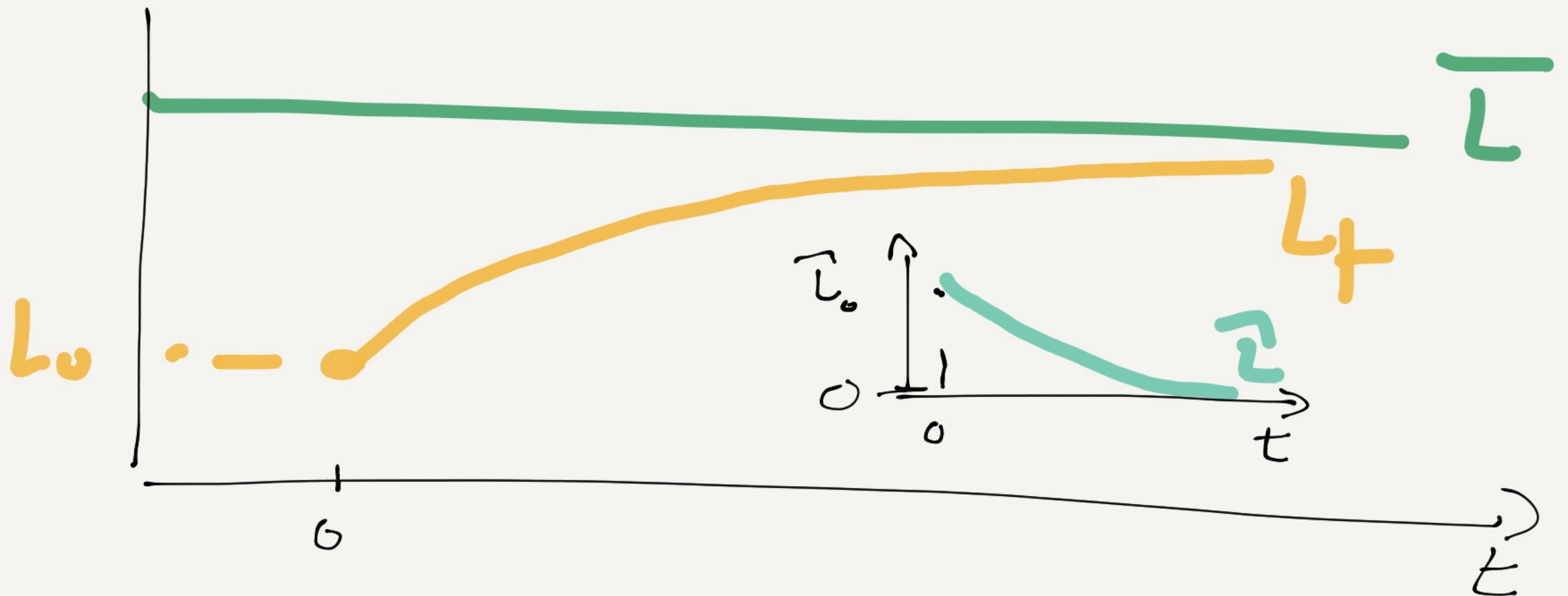
$$L_2 = (1 - \lambda_1) \bar{L} + \lambda_1 L_1 = (1 - \lambda_1)(1 + \lambda_1) \bar{L} + \lambda_1^2 L_0$$

$$L_3 = (1 - \lambda_1) \bar{L} + \lambda_1 L_2 = (1 - \lambda_1)(1 + \lambda_1 + \lambda_1^2) \bar{L} + \lambda_1^3 L_0$$

$$L_t = \bar{L}(1-\lambda_1) \sum_{j=0}^{t-1} \lambda_1^j + \lambda_1^t L_0$$

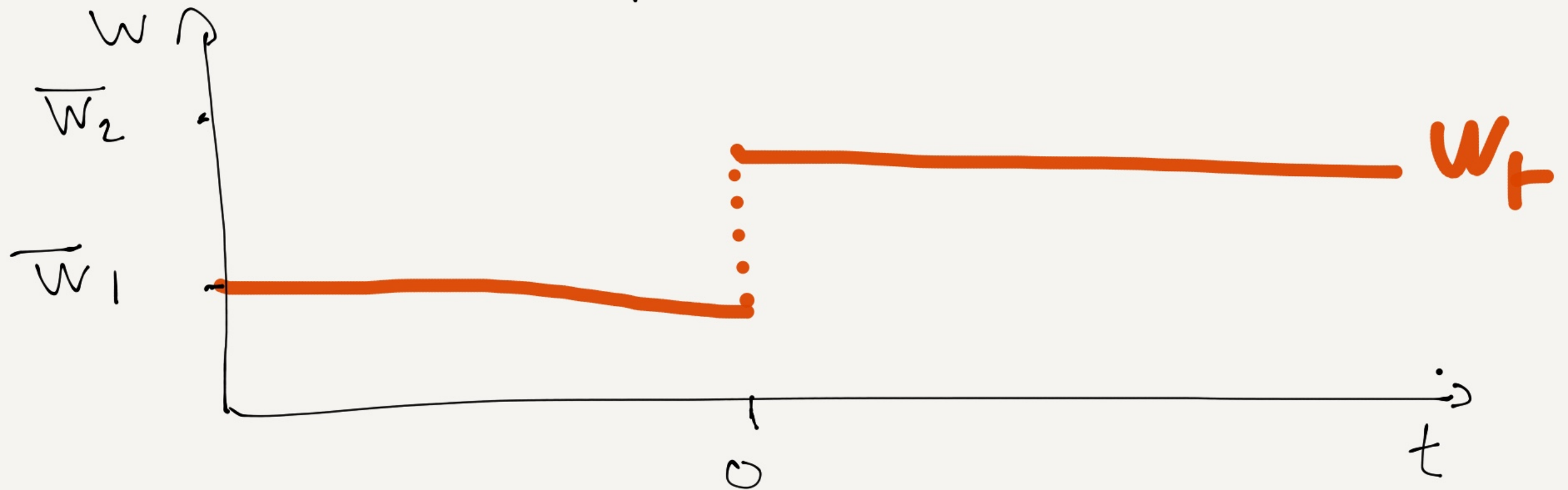
$$\lim_{t \rightarrow \infty} L_t = (1-\lambda_1) \bar{L} \frac{1}{1-\lambda_1} + 0$$

$$= \bar{L}$$

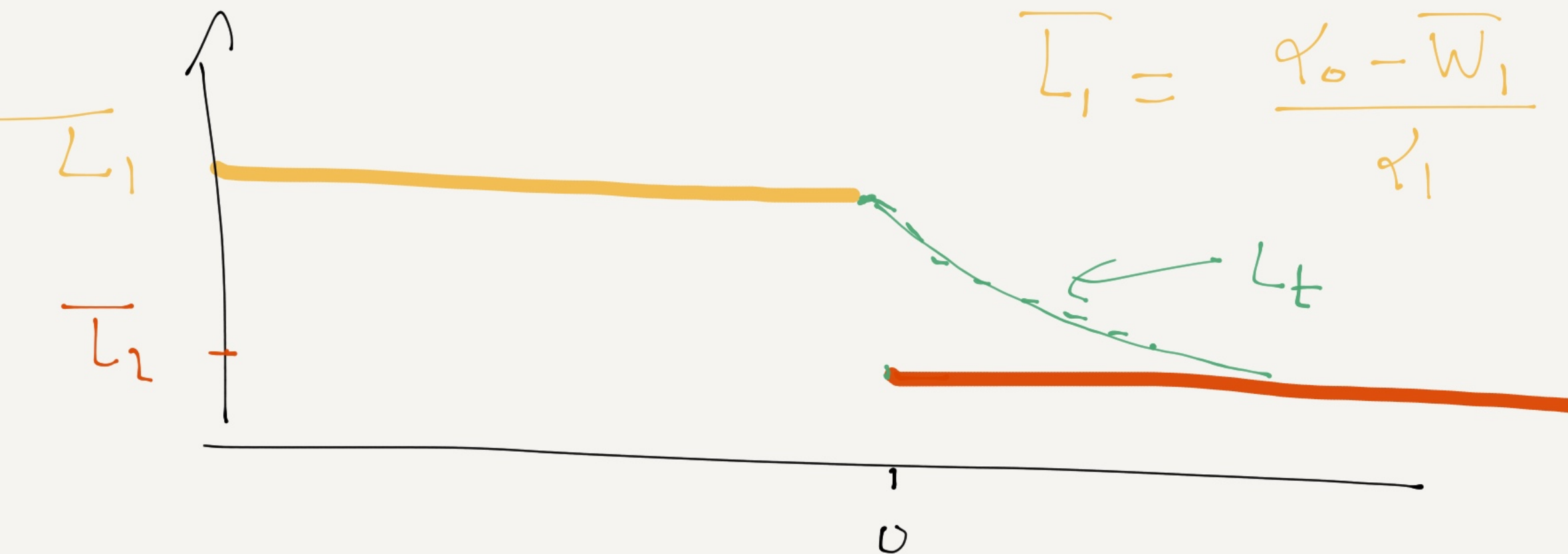


Experiment

- from $-\infty$ to -1 , $w_t = \overline{w}_1$
- $\overline{w}_t$ is expected to stay at $\overline{w}_1$ for even
- Unexpectedly, in 0, $w_0 = \overline{w}_2 > \overline{w}_1$
- This will look for even.



• Assume $L_t = \bar{L}_1$ for $t < 0$



$$\bar{L}_2 = \frac{q_0 - \bar{W}_2}{q_1} < \bar{L}_1$$

$$L_0 = (1-\lambda_1)\overline{L}_2 + \lambda_1 L_{-1} = (1-\lambda_1)\overline{L}_2 + \lambda_1 \overline{L}_1$$

$$L_1 = (1-\lambda_1)\overline{L}_2 + \lambda_1 L_0$$

$$= (1-\lambda_1)\overline{L}_2 + \lambda_1(1-\lambda_1)\overline{L}_2 + \lambda_1^2 \overline{L}_1$$

$$= (1-\lambda_1)(1+\lambda_1)\overline{L}_2 + \lambda_1^2 \overline{L}_1$$

⋮

$$L_t = \overline{L}_2 \left((1-\lambda_1) \sum_{j=0}^{t-1} \lambda_1^j \right) + \lambda_1^t \overline{L}_1$$

$$\lim_{t \rightarrow \infty} L_t = \overline{L}_2$$

